

Open Source Tools for Agentic AI: An Overview

This outline of the essential tools, platforms, and frameworks across each layer of the agentic AI stack provides a structured blueprint for organisations seeking to adopt these systems at scale.



Agentic AI refers to AI systems capable of taking actions, using tools, coordinating workflows, and achieving goals with minimal human intervention. Unlike traditional generative AI, which focuses on producing text, images, or code, agentic AI systems can:

- Understand objectives
- Break them into tasks
- Select tools or APIs
- Execute multi-step workflows
- Monitor progress
- Adapt based on feedback
- Collaborate with other agents or humans

Key characteristics of agentic AI

Agentic AI systems plan, decide, act, and learn. This is fundamentally different from conversational or generative AI because they don't just produce responses; they autonomously

execute tasks and drive outcomes across systems. The process involves perception, reasoning, planning, action and reflection.

Perception: Agentic AI gathers information from its surroundings and different sources like sensors, databases, and user/digital interfaces. This involves analysing text, images, or other forms of data to understand the situation.

Reasoning: It analyses the gathered data to understand the context, identify relevant information, and formulate potential solutions using large language models (LLMs).

Planning: The agent uses the information it gathered to develop a plan. This involves setting goals, breaking them down into smaller steps and charting a path to achieve goals.

Action: The agent executes the plan by interacting with systems to perform the tasks and make decisions.

Reflection: The agent evaluates outcomes and adjusts future behaviour. This reflection loop improves accuracy and reduces manual review over time.